INCH-POUND
MIL-PRF-3098/40F
9 December 2004
SUPERSEDING
MIL-PRF-3098/40E
27 August 1997

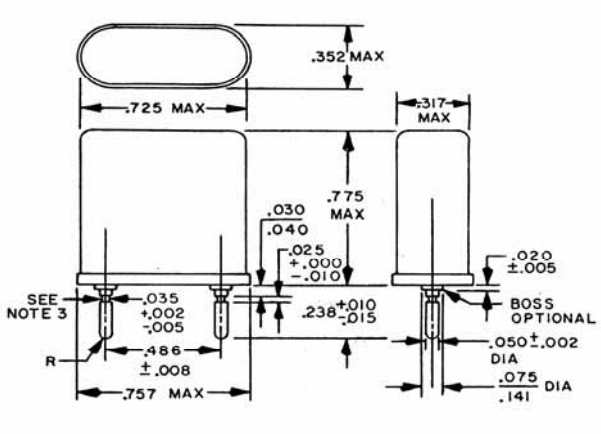
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR62/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental: controlled; antiresonance.



Inches	mm	Inches	mm
.002	0.05	.050	1.27
.005	0.13	.075	1.91
.008	0.20	.141	3.58
.010	0.25	.238	6.05
.015	0.38	.317	8.05
.020	0.51	.352	8.94
.025	0.64	.486	12.34
.030	0.76	.725	18.42
.035	0.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR62/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Frequency tolerances:

Operating temperature range: ±10 parts per million (ppm).

Room temperature: : ±70 ppm.

Frequency stability: ±5 ppm.

Equivalent resistance: See table I.

Antiresonance, load capacitance: 32.0 pF ±0.2 pF.

Mode of oscillation: Fundamental.

Reference temperature: +75°C ±1°C.

Temperature ranges:

Operable: -55°C to +70°C and +80°C to +90°C, inclusive.

Operating (controlled): +70°C to +80°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Aging:

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Frequency change permitted: ±5 ppm.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.80 to 0.90	600
0.90+ to 1.00	570
1.00+ to 1.12	540
1.12+ to 1.25	490
1.25+ to 1.37	450
1.37+ to 1.50	410
1.50+ to 1.62	370
1.62+ to 1.75	330
1.75+ to 1.87	300
1.87+ to 2.00	290
2.00+ to 2.12	270
2.12+ to 2.25	240
2.25+ to 2.60	190
2.60+ to 3.00	150
3.00+ to 3.40	110
3.40+ to 3.75	90
3.75+ to 4.00	75
4.00+ to 5.00	60
5.00+ to 7.00	35
7.00+ to 10.00	24
10.00+ to 15.00	22
15.00+ to 20.00	20

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - CR

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DLA - CC

(Project 5955-0776)

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